

Scientific Machine Learning for Computational Problems

- Overview of optimal SciML methods
- Applied to real-world engineering and physics
- Covers computational mechanics to biophysical modeling

Challenge Domain

Fick's Diffusion with Algebraic Constraints

Core Challenge: Solving coupled Fick's diffusion PDEs with algebraic constraints, requiring satisfaction of both differential and algebraic conditions.

PINNs: Direct Coupled Solution

- Adds two residual terms to loss function: $\mathcal{L}_{PDE}$ for Fick's law and $\mathcal{L}_{alg}$ for algebraic constraint.
- Network trained to satisfy both simultaneously through multi-objective optimization.

FNO/Poseidon: Evolution Operator

- Learns an operator mapping initial concentration fields to final states.
- Uses PINN or traditional solver datasets.
- Implicitly captures combined diffusion and constraint effects without explicit enforcement.

Implementation Considerations

- **PINNs:** Critical to balance weights between PDE and algebraic loss terms. Consider adaptive weighting or gradient-based methods for convergence.
- **FNO:** Ensure training data covers diverse constraint parameter regimes.

Challenge Domains

Linear Elastic Cantilever Beam Analysis

Challenge: Solve static elasticity for displacement and stress fields without mesh generation.

- **Energy-Based PINNs**

Minimize Total Potential Energy (Π) instead of strong-form PDE for mesh-free displacement solutions.

- **FNO/DeepONet Surrogates**

Map load functions or geometry parameters to solution fields for rapid design and optimization.

Challenge Domain

Recovering Pendulum Dynamics from Data

Core Challenge: Discover governing equations and parameters for pendulum motion from trajectory data, merging data-driven learning with physics discovery.

Parameter Discovery with PINNs

- **Approach:** Parameter identification (not full equation discovery).
- **Strategy:** Assume equation form $\ddot{\theta} + k \sin(\theta) = 0$. Make 'k' a trainable parameter.
- PINN loss combines data mismatch and PDE residual to find $k = g/L$.
- **Use Case:** Effective when the functional form is known, but coefficients are uncertain.

Black-Box Modeling

- **Method:** DeepONet / FNO for surrogate dynamics.
- **Strategy:** Learn operators mapping initial conditions to complete trajectories.
- Creates accurate simulators without explicit interpretable equations.
- **Use Case:** Useful for prediction and control, but does not reveal underlying physical principles.

Challenge Domain

Anisotropic Wave Equation Inverse Problem

Core Challenge: Recover unknown, spatially-varying density $\rho(x,y)$ from sparse sensor measurements of wave propagation in anisotropic media.

The governing equation is:

$$\rho \ddot{u} = \nabla \cdot (C : \nabla u)$$

This inverse problem is ill-posed, requiring regularization with physics constraints and careful optimization strategies.

Method Selection Strategy

1 PINNs: Inverse Problem

- Treat unknown density $\rho(x,y)$ as trainable parameters or a neural network.
- Minimize combined loss: PDE residual + data mismatch with observations u_{obs} .

2 FNO/Poseidon: Forward Surrogate

- Train on datasets mapping material properties (ρ, C) to resulting wavefields $u(x,t)$.
- Essential for rapid forward simulations in iterative inversion schemes.

Challenge Domain

Constitutive Models for Soft Biological Tissue

Core Challenge: Develop data-driven Constitutive Artificial Neural Networks (CANNs) to model nonlinear, anisotropic, and viscoelastic biological tissues.

Focus: Learn thermodynamically consistent strain energy functions (Ψ) while maintaining material objectivity and stability.

PINNs: Core CANN Framework

- Learn strain energy density (Ψ) as a neural network.
- Physics loss enforces $P = \partial \Psi / \partial F$ (stress-deformation).
- Experimental data provides data loss for material parameter identification.

DeepONet: History-Dependent Response

- Model viscoelastic behavior.
- Learn operators mapping strain history to current stress states.
- Captures time-dependent relaxation and creep.

FNO: Full-Field Surrogate

- Learn mappings from displacement fields to stress fields.
- Useful for inverse characterization from full-field experimental data.

Challenge Domain

Eddy Current Inspection via Maxwell's Equations

Core Challenge: Simulate eddy currents for non-destructive testing (NDT) and detect defects from sensor impedance measurements.

01

PINNs: Forward & Inverse

- **Forward:** Embed Maxwell's equations as physics loss to solve for E/B fields.
- **Inverse:** Treat conductivity (σ) as a trainable field. Optimize for defect location/characteristics using sensor data.

02

FNO/Poseidon: Real-Time Inspection

- Train on datasets mapping defect shapes/locations to sensor signals.
- Provides ultra-fast forward surrogate, crucial for real-time industrial inspection.

03

DeepONet: Excitation-to-Signal Operator

Learn the operator mapping coil excitation parameters (frequency, position, current) to impedance signatures for rapid scan analysis.

Challenge Domain

3D Plant Reconstruction & Growth Modeling

Core Challenge: Modeling complex 3D plant growth dynamics from sparse, noisy sensor data while integrating biophysical principles and environmental factors.

PINNs

Best for: Enforcing biophysical growth laws (PDEs for nutrient transport) and physics-based constraints with sparse data.

FNO

Best for: Fast 3D structure prediction from low-resolution sensor inputs or high-fidelity simulations.

DeepONet

Best for: Environmental response mapping; modeling how input functions (e.g., light, humidity) affect growth rates and morphology.

Aurora / Poseidon

Best for: Transfer learning enhancement; leveraging pre-trained physics models for improved generalization in nutrient and fluid flow.

Challenge Domain

Elastoplastic Soil Mechanics

Core Challenge: Model highly nonlinear, path-dependent soil stress-strain responses using Mohr-Coulomb or Drucker-Prager constitutive models.

PINNs for Forward Problem

Application: Constitutive ANNs (CANNs)

- Embed equilibrium, kinematic, and constitutive laws as loss terms.
- Solve for displacement and stress fields under given boundary conditions.

DeepONet for Constitutive Modeling

Application: Data-driven material response

- Learn operator mapping strain history to stress response.
- Utilizes experimental or PINN-generated data.

FNO for Full-Field Solutions

Application: Fast surrogate solver

- Train on datasets mapping boundary conditions to stress-strain fields.
- Enables rapid evaluation for design optimization.

Challenge Domain

Hybrid Motion Modeling

Core Challenge: Model particle/robot motion by integrating known physics with learned forces from data.

Gray-Box PINNs

Split dynamics into known physics (F_{known}) and unknown forces (F_{unknown}). Represent F_{unknown} as a neural network.

DeepONet for Environments

Learn operators mapping environment representations (e.g., obstacle fields) to particle trajectories across diverse scenarios.

Aurora/Poseidon Fine-Tuning

Accelerate training by using pre-trained continuum dynamics knowledge as a prior for the F_{unknown} network.